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Editorial Office
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Dear Editor,

We are pleased to submit our manuscript, "Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning", for consideration as a regular article in Data & Knowledge Engineering.

The manuscript fits Data & Knowledge Engineering because it addresses a core knowledge-engineering problem: how exposed intermediate knowledge artifacts can be represented, organized, and maintained when human curation budgets are limited. Modern knowledge-intensive systems expose reasoning traces, retrieval checks, entity bindings, graph nodes, verification records, and process annotations, but these artifacts are often available only as heterogeneous signals rather than reusable maintenance records.

We introduce Structurally-Calibrated Functional Attribution (SC-FMA) as a knowledge representation and transformation layer for this setting. SC-FMA converts observable intermediate artifacts into structured audit records that preserve supplied annotation or utility fidelity while adding graph-aware structural-dependency, redundancy, bottleneck, audit-reason, and maintenance-action fields. The novelty lies in the structured representation mechanism that turns intermediate artifacts into maintainable knowledge objects.

The contribution to knowledge maintenance is practical. The resulting records support fixed-budget knowledge audit, knowledge curation, redundancy consolidation, dependency inspection, weak-signal repair, and reuse of curated artifacts across the knowledge lifecycle. The manuscript evaluates this representation behavior through PRM800K annotation evidence, a Countries-KG graph-representation study, rule-derived audit-record construction, and controlled calibration analysis, without changing the reported mathematical formulation or empirical results.

The reported validation is limited to observable intermediate artifacts and fixed-budget audit settings; deployed maintenance studies and external task generalization require separate evaluation.

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Sincerely,

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